

Retail Store Sales Analysis — Auto Analysis

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Executive Summary

This report summarizes the dataset quality, exploratory findings, and model performance for project "Retail Store Sales Analysis".

Dataset Overview

File	retail_sales_dataset.csv
Rows	3000
Columns	14
Missing Values	0
Duplicate Rows	0
Cleaned	True

Modeling & Evaluation

Model Type	xgboost
Problem Type	classification
Target Column	store_city

Metrics

Metric	Value
accuracy	0.245
precision	0.2165
recall	0.245
f1	0.2263
roc_auc	0.5033

confusion_matrix	{'labels': ['Faisalabad', 'Islamabad', 'Karachi', 'Lahore', 'Multan'], 'matrix': [[4, 19, 35, 18, 6], [8, 19, 48, 44, 7], [15, 32, 73, 56, 2], [16, 28, 52, 49, 8], [4, 13, 29, 13, 2]]}
class_distribution	{'Karachi': 178, 'Lahore': 153, 'Islamabad': 126, 'Faisalabad': 82, 'Multan': 61}
cv_scores	[0.20935231221353945, 0.22860532144368967, 0.23234691262947937, 0.22640106201939889, 0.2304491649620823]
cv_mean	0.2254
baseline	{'baseline_score': 0.1357, 'baseline_metric': 'f1', 'model_score': 0.2263, 'improvement': 0.0906, 'cv_std': 0.0083, 'clearly_beats_baseline': True}
leakage_warnings	[]
post_outcome_excluded	[]
post_outcome_included	[]

Feature Importance

Feature	Importance
product_category_Sports	0.0562
returned_No	0.0494
unit_price	0.0483
product_category_Clothing	0.0483
product_category_Groceries	0.048
product_category_Home & Kitchen	0.0475
sales_channel_In-Store	0.0472
sales_channel_Mobile App	0.0466
sales_channel_Online	0.0465
profit	0.0463
quantity	0.0456
sales_amount	0.0455

order_date_month	0.045
order_date_dayofweek	0.0438
product_category_Beauty	0.0432
order_date_year	0.0431
promotion_applied_No	0.0431
customer_rating	0.0427
discount_percent	0.0425
customer_segment_Premium	0.0422

AI Business Insight [AI-generated insight](#)

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Limitations

Model predictions are estimates based on historical data and may contain uncertainty. AI-generated text explains verified analytical results but does not independently calculate metrics. Always validate significant business decisions with domain expertise.